

Executive Summary

TNSVT V2 es una **plataforma SaaS de trading algorítmico** de nivel empresarial que integra un motor de señales multi-broker, copy trading 1:N configurable por cuenta, inteligencia artificial autoalojada (Ollama), y un framework de riesgo en tiempo real — todo sobre una arquitectura orientada a eventos con NATS JetStream.

13 microservicios en producción escritos en Go y Python, desplegados con un solo `docker-compose up`, desde un solo servidor hasta un cluster Kubernetes HA.

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Problem & Vision

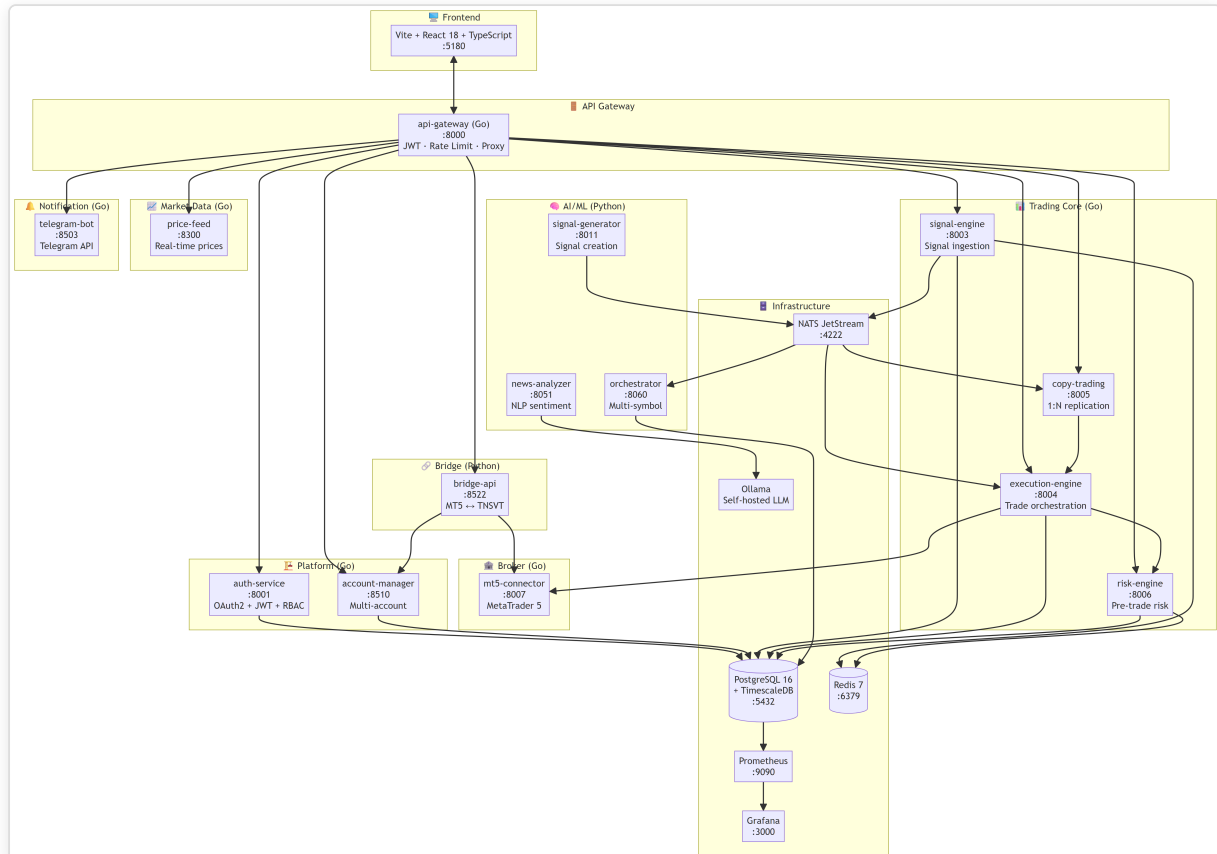
Current Limitations (V1)

Dimensión	V1 (Legacy)	V2 (Objetivo)
Stack	PHP Symfony monolítico + FastAPI aislado	Go microservicios + Python AI/ML
Mensajería	Sin bus centralizado	NATS JetStream event sourcing
Base de datos	SQLite (sin escalabilidad)	PostgreSQL + TimescaleDB
Usuarios concurrentes	~50	100,000+
Brokers	1 (MT5)	5+ (MT5, cTrader, Binance, Bybit, IBKR)
Latencia de ejecución	2-5 segundos	< 100ms (p99)
Copy trading	Rígido 1:1, sin configuración	1:N configurable por cuenta
AI/ML	Sin AI productivo	Ollama self-hosted + RAG + scoring NLP
Observabilidad	Sin métricas	OpenTelemetry + Prometheus + Grafana
Despliegue	Manual, frágil	Docker Compose → Swarm → K8s

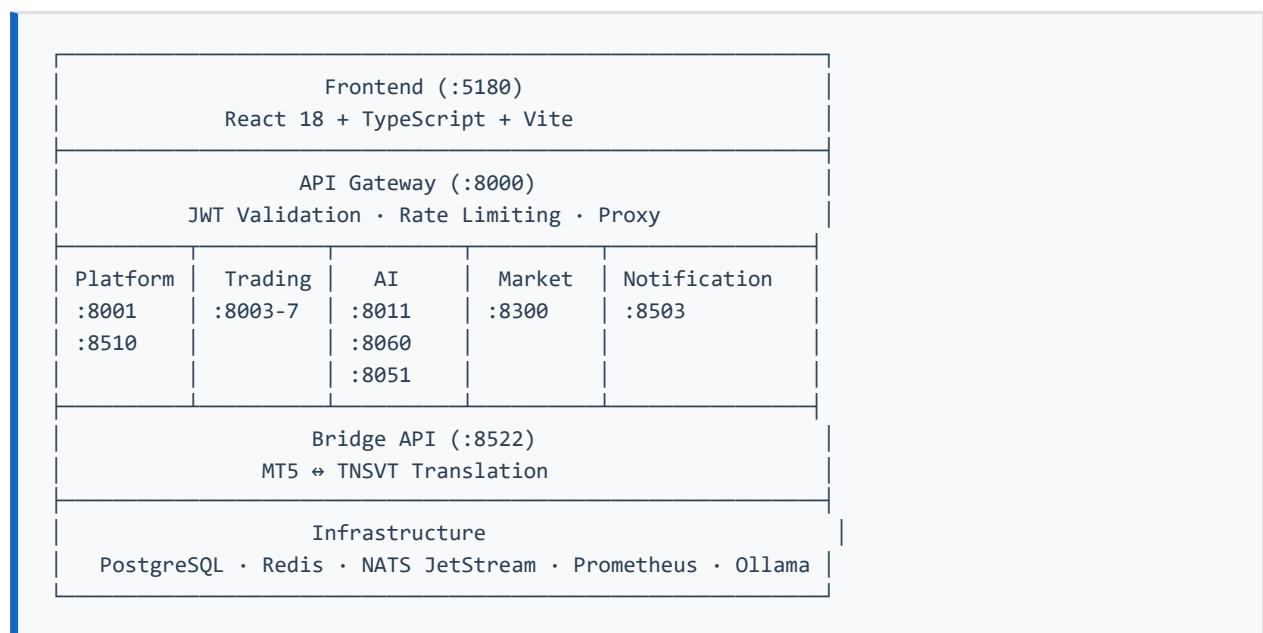
Pilares Arquitectónicos

1. **Event-Driven:** NATS JetStream como backbone de mensajería
 2. **Desacoplamiento:** Cada servicio es independiente, desplegable y escalable
 3. **AI-First:** Modelos de lenguaje autoalojados para scoring y análisis
 4. **Multi-Tenancy:** Schema-per-tenant PostgreSQL desde el diseño
 5. **Autoalojable:** Sin dependencias de APIs externas para operación core
-

Architecture Overview



Layer Architecture



Technology Stack

Component	Technology	Justification
Backend (Core)	Go 1.22+	Concurrencia nativa, binarios estáticos, rendimiento predecible
Backend (AI/Bridge)	Python 3.12+	Ecosistema ML/NLP (transformers, langchain)
Frontend	React 18 + Vite + TypeScript	CSR puro, despliegue CDN, sin SSR
Message Bus	NATS JetStream	Persistente, alta velocidad, clustering nativo
Database	PostgreSQL 16 + TimescaleDB	Relacional + series temporales
Cache	Redis 7	Rate limiting, sesiones, señal dedup
AI	Ollama (self-hosted LLM)	Sin dependencia cloud, privacidad total
Monitoring	Prometheus + Grafana	Stack estándar open-source
Container	Docker + Docker Compose	Despliegue reproducible

Services Catalog

Gateway

api-gateway (:8000)

Rol: Proxy reverso, JWT validation, rate limiting, entry point único.

Tech: Go 1.22 + chi router + JWT middleware

Routes:

Route	Service	Auth
/api/v1/auth/*	auth-service	No
/api/v1/me	auth-service	JWT
/api/v1/signals/*	signal-engine	JWT
/api/v1/accounts/*	account-manager	JWT
/api/v1/copy/*	copy-trading	JWT
/api/v1/execute/*	execution-engine	JWT
/api/v1/risk/*	risk-engine	JWT
/api/v1/prices/*	price-feed	JWT
/api/v1/bridge/*	bridge-api	JWT
/api/v1/notify/*	telegram-bot-service	API Key
/api/v1/orchestrator/*	orchestrator	JWT

Platform Services

auth-service (:8001)

Rol: Autenticación OAuth2, JWT emisión/refresh, RBAC, gestión de usuarios.

Endpoints:

- `POST /api/v1/auth/login` — Email + password → JWT pair
- `POST /api/v1/auth/refresh` — Refresh token → new JWT pair
- `GET /api/v1/auth/me` — Current user profile + permissions
- `POST /api/v1/auth/logout` — Invalidate session
- `GET /api/v1/auth/users` — List users (admin)
- `POST /api/v1/auth/users` — Create user (admin)

DB: `users`, `roles`, `permissions`, `refresh_tokens`

account-manager (:8510)

Rol: Gestión de cuentas MT5 multi-broker, vinculación bridge.

Endpoints:

- `GET /api/v1/accounts` — List accounts
- `POST /api/v1/accounts` — Register account
- `GET /api/v1/accounts/{id}` — Account details

Trading Core

signal-engine (:8003)

Rol: Ingesta, validación, deduplicación y almacenamiento de señales.

Flujo:

1. Recibe señal (NATS o HTTP)
2. Valida formato y broker
3. Deduplica por symbol+timestamp
4. Almacena en PostgreSQL
5. Publica en NATS `trading.signal.validated`

execution-engine (:8004)

Rol: Orquestación de ejecución de órdenes con retry y backoff.

Endpoints:

- `POST /api/v1/execute` — Execute order (with risk check)
- `GET /api/v1/execute/{id}` — Order status
- `GET /api/v1/execute` — List orders

Resiliencia: 2 retries, 30s timeout, circuit breaker per broker

copy-trading (:8005)

Rol: Replicación 1:N de señales a múltiples cuentas.

Capacidades:

- Matching por grupos predefinidos
- Escalado de lote por cuenta
- Filtros por símbolo, riesgo, horario
- Creación de CopyJob por cada order

risk-engine (:8006)

Rol: Pre-trade risk validation en tiempo real.

Checks:

- Max drawdown (< 15%)
- Max open positions (< 3)
- Per-symbol exposure limits
- Account-level daily loss limit

AI / ML

signal-generator (:8011)

Rol: Generación programática de señales basada en reglas y data de mercado.

orchestrator (:8060)

Rol: Orquestación multi-symbol de señales generadas por AI.

news-analyzer (:8051)

Rol: Análisis NLP de noticias financieras vía Ollama.

Flujo: Fetch news → Ollama LLM analysis → Sentiment score → NATS signal

Market Data

price-feed (:8300)

Rol: Streaming de precios en tiempo real desde múltiples fuentes.

WebSocket: Actualizaciones sub-second de bids/asks

Notification

telegram-bot-service (:8503)

Rol: Proxy de notificaciones a Telegram con soporte multi-chat.

Endpoints:

- `GET /api/v1/notify/health` — Health check (proxy-aware)
- `POST /api/v1/notify/send` — Send message to chat

Bridge

bridge-api (:8522)

Rol: Traductor MT5 ↔ TNSVT, sincronización de cuentas y señales.

Flujo:

1. MT5 Bot envía señal → bridge-api
2. bridge-api enruta a signal-engine
3. bridge-api consulta account-manager para mapping
4. bridge-api envía órdenes a MT5

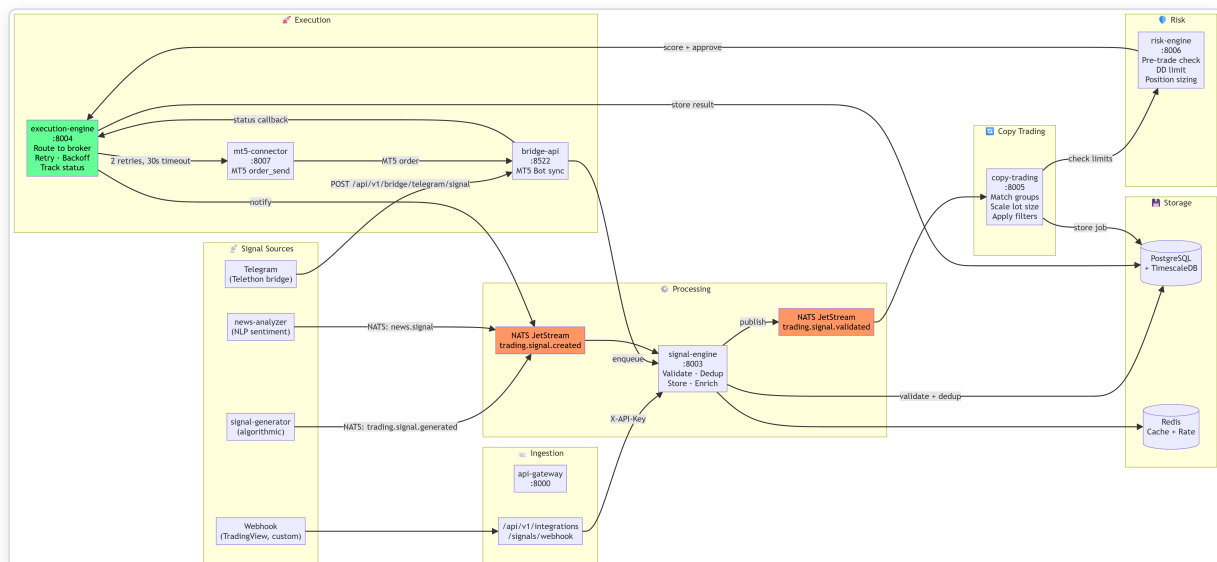
Broker

mt5-connector (:8007)

Rol: Conexión nativa con MetaTrader 5, envío de órdenes vía API.

Data Flows

End-to-End Signal Pipeline



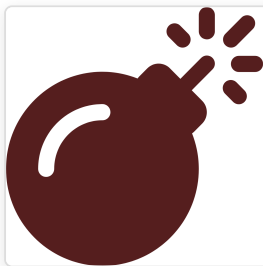
Flow completo:

1. **Source:** Una señal llega desde signal-generator (algorítmico), webhook (TradingView), Telegram bridge, o news-analyzer (NLP)
2. **Ingestion:** api-gateway recibe y reenvía a signal-engine
3. **Validation:** signal-engine valida formato, simbolo, deduplica
4. **Distribution:** Publica en NATS `trading.signal.validated`
5. **Copy Trading:** copy-trading consume, escala lotes, crea CopyJob
6. **Risk Check:** risk-engine valida drawdown, posiciones, exposición
7. **Execution:** execution-engine envía a broker con retry+backoff
8. **Broker:** mt5-connector ejecuta en MT5 y retorna status

Tiempos (benchmark):

Etapa	Latencia esperada
signal → NATS	< 5ms
NATS → signal-engine	< 10ms
copy-trading	< 50ms
risk-engine	< 30ms
execution-engine → broker	< 100ms
Total (p99)	< 200ms

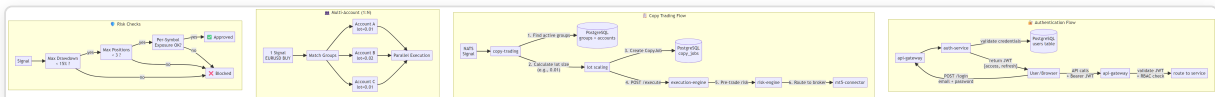
Service Dependencies



Syntax error in text

mermaid version 10.9.6

Key Data Flows



Authentication Flow

User → POST /login → api-gateway → auth-service → PostgreSQL → JWT tokens
User → API call + Bearer JWT → api-gateway → validate JWT → route to service

Copy Trading 1:N

```
1 Signal (EURUSD BUY) → Match Groups → Account A (lot=0.01)
                                   → Account B (lot=0.02)
                                   → Account C (lot=0.01)
                                   → Parallel Execution
```

Risk Gate Flow

Signal → Max Drawdown < 15%? → Max Positions < 3? → Per-Symbol Exposure OK? → Approved
↓
Blocked ❌

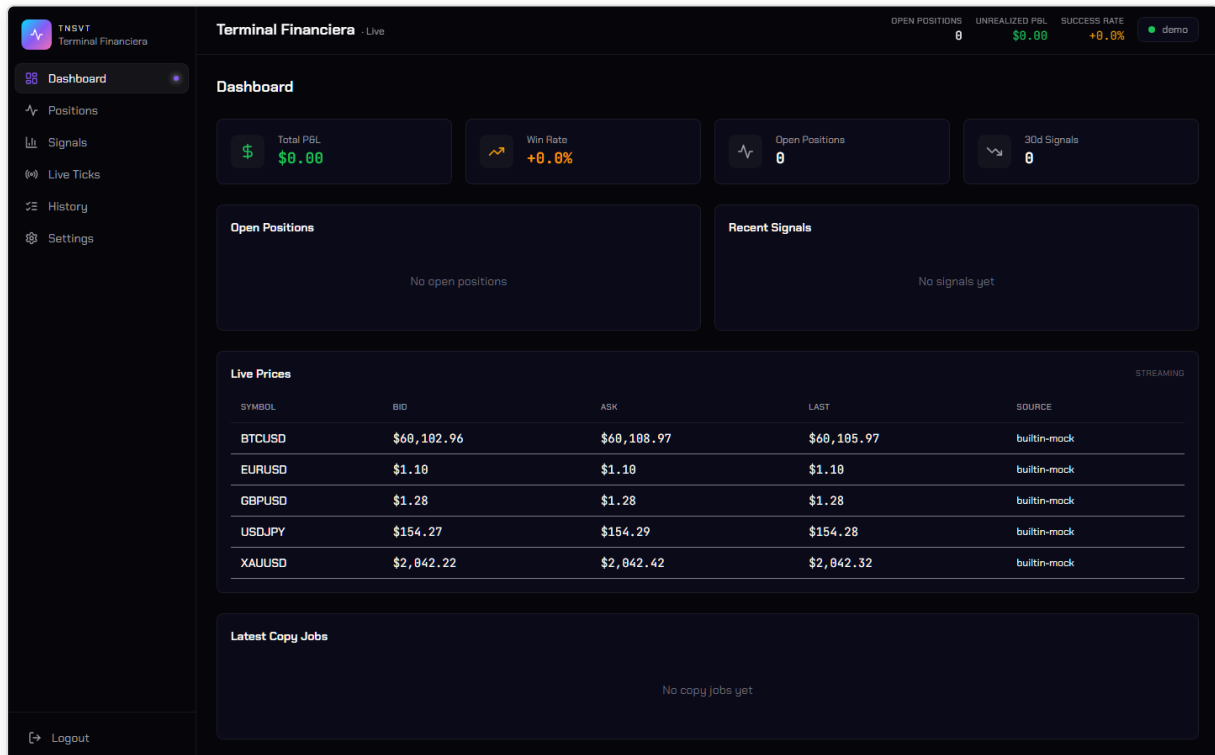
Infrastructure & Deployment

Container Architecture

Todos los servicios se despliegan como contenedores Docker:

```
# docker-compose.yml (core)
services:
  nats:      image: nats:2.10 # JetStream enabled
  postgres:  image: postgres:16 # + timescaledb
  redis:     image: redis:7
  gateway:   build: apps/gateway/api-gateway
  auth:      build: apps/auth-service
  signal-gen: build: apps/ai/signal-generator
# ... 13 services total
```

Ciclo de Vida de Señal



Estados de un CopyJob

```
PENDING → ACTIVE → COMPLETED
      ↓
    FAILED (retry 2x)
      ↓
    FAILED (final)
```

Credenciales

- **Admin Panel:** `http://localhost:8000`
 - **Login:** `admin@tnsvt.local` / `Admin123!Demo`
 - **PostgreSQL:** `localhost:5432` / `tnsvt` / `tnsvt_trading`
 - **NATS:** `localhost:4222`
 - **Redis:** `localhost:6379`
-

AI & Machine Learning

Ollama Integration

El modelo de lenguaje se ejecuta **self-hosted** vía Ollama, sin dependencia de APIs externas. Esto garantiza:

- **Privacidad total:** datos de trading nunca salen de la infraestructura
- **Latencia controlada:** ~500ms por inferencia en GPU local
- **Costo predecible:** sin costos API por token
- **Offline-first:** opera sin conexión a internet

news-analyzer Pipeline

```
News Article → Fetch HTML → Clean Text → Ollama LLM → Sentiment Score  
                                                    (BUY/SELL/HOLD + confidence)  
                                                    ↓  
NATS: news.signal  
                                                    ↓  
signal-engine
```

Modelos Utilizados

Modelo	Propósito	Tamaño
Llama 3.1 8B	Análisis de sentimiento financiero	4.7GB
Mistral 7B	Scoring de señales (fallback)	4.1GB

Observability & Resilience

Monitoring Stack

Component	Rol	Puerto
Prometheus	Metrics collection	:9090
Grafana	Visualization	:3000
NATS JetStream	Event sourcing + persistence	:4222

Resilience Patterns

Retry with Exponential Backoff

Cada llamada a broker implementa:

- **Max retries:** 2
- **Timeout:** 30s
- **Backoff:** 1s, 5s (exponencial + jitter)

Circuit Breaker

Si un broker falla 3 veces consecutivas, el circuito se abre por 60s.

Graceful Degradation

- Si Ollama falla → fallback a scoring basado en reglas
 - Si PostgreSQL falla → Redis cache como respaldo de solo lectura
 - Si broker falla → cola de reintentos en NATS (no pérdida de orden)
-

Security

Authentication & Authorization

- **JWT:** Access token (15min) + Refresh token (7 días)
- **RBAC:** Roles `admin`, `trader`, `viewer` con permisos granulares
- **Rate limiting:** 100 req/min por IP, 1000 req/min por API key

Network Security

- **Gateway como único entry point:** todos los servicios internos sin exposición
- **Proxy-aware:** telegram-bot-service detecta X-Forwarded-* headers
- **PostgreSQL:** autenticación md5, conexiones solo desde red interna Docker

Data Protection

- **Passwords:** bcrypt (cost=12)
 - **JWT Secrets:** 256-bit random, rotables
 - **API Keys:** hasheadas en DB, prefix público para identificación
-

Roadmap

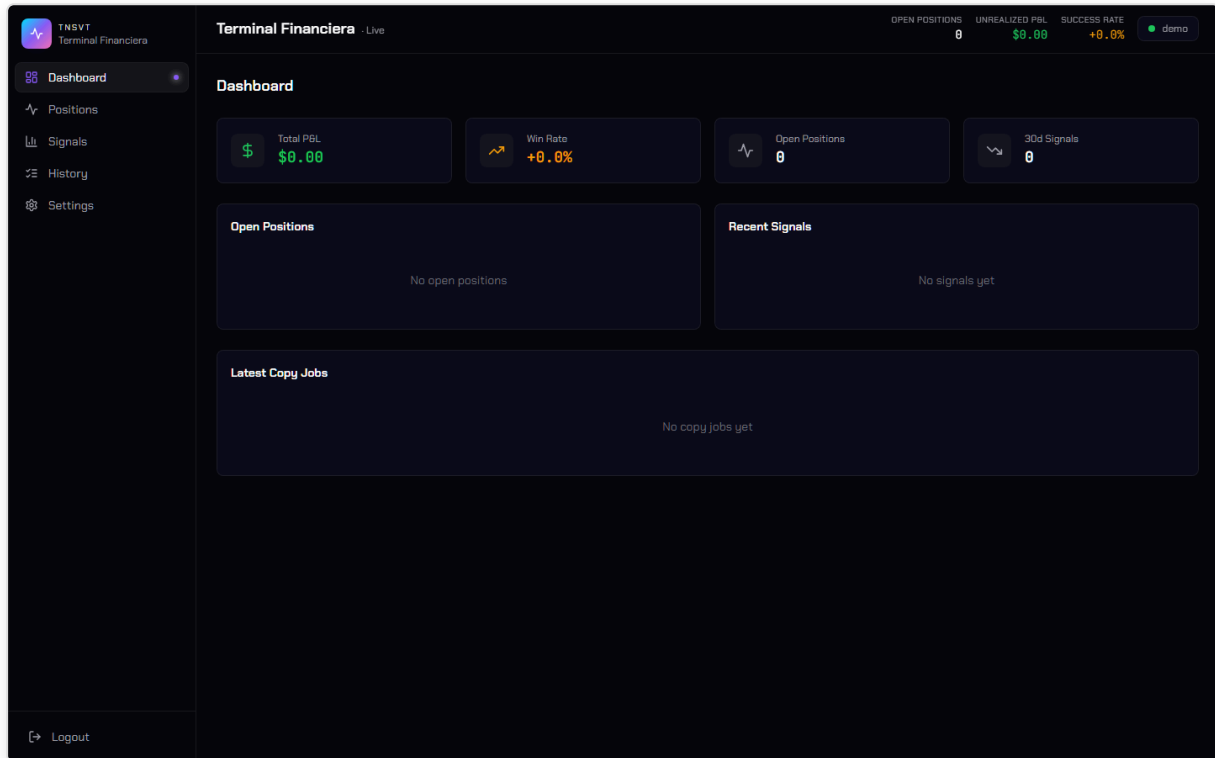
Fase	Q	Features
1 (Actual)	Q3 2026	Core trading + copy + risk + bridge MT5
2	Q3 2026	Multi-broker (cTrader), WebSocket streaming, Grafana dashboards
3	Q4 2026	Frontend React completo, Tauri desktop app, multi-tenancy
4	Q1 2027	Scale to 100k users, Kubernetes HA, white-label

Phase 2 (Next)

- ☐ Multi-broker: cTrader connector
 - ☐ WebSocket streaming bidireccional
 - ☐ Grafana dashboards operacionales
 - ☐ Sistema de alertas vía WebSocket
 - ☐ Backtesting engine
 - ☐ Smart Order Routing
-

Appendix: Screenshots

Dashboard — Trading Overview



Active Positions

TNSVT
Terminal Financiera

Dashboard

Positions

Signals

History

Settings

Logout

Terminal Financiera · Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Positions

Open (0)

No positions

Closed (0)

No positions

Signal History

TNSVT
Terminal Financiera

Dashboard

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Terminal Financiera · Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Signals

No signals received yet

Execution History

TNSVT
Terminal Financiera

Dashboard

Positions

Signals

History

Settings

Terminal Financiera · Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Trade History

No closed trades yet

Logout

Settings & Configuration

TNSVT
Terminal Financiera

Dashboard

Positions

Signals

History

Settings

Terminal Financiera · Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Settings

Account

User ID00000000-0000-0000-0000-000000000001

Emaildemo@tnsvt.com

Usernamedemo

RoleADMIN

Tenant ID00000000-0000-0000-0000-000000000001

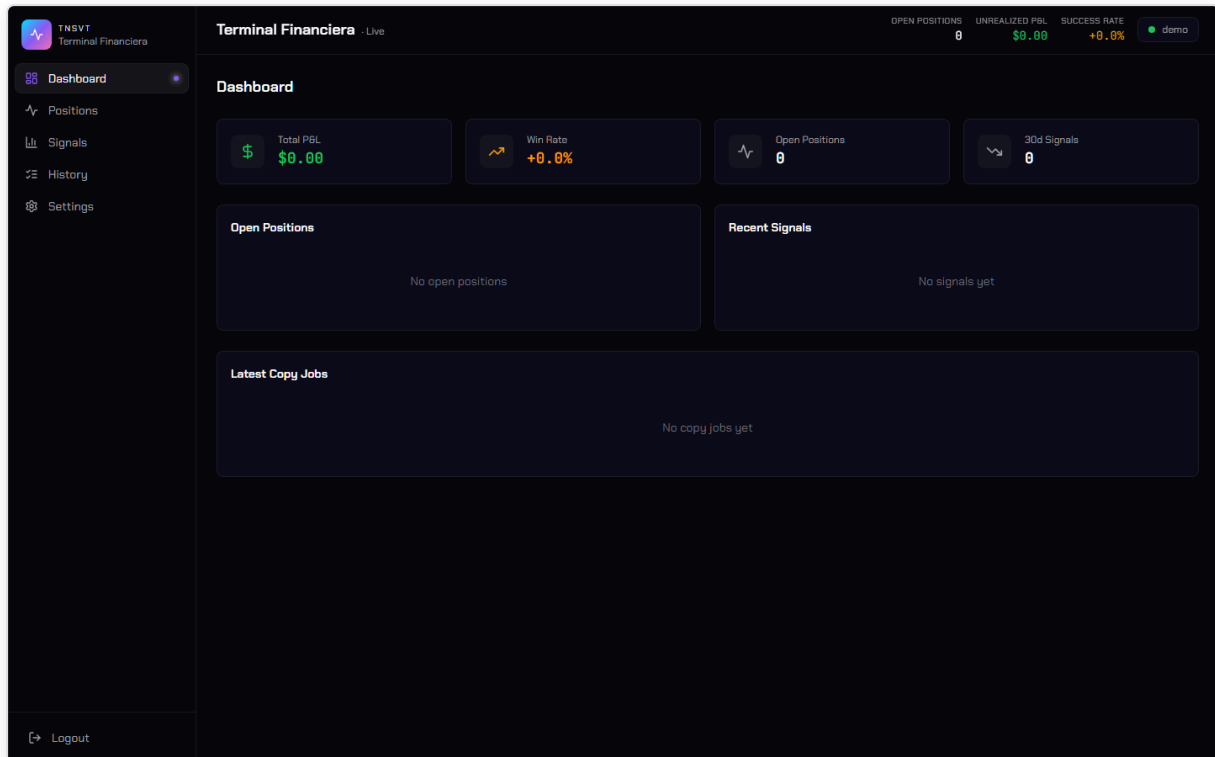
API Access

Use your JWT token for API authentication.

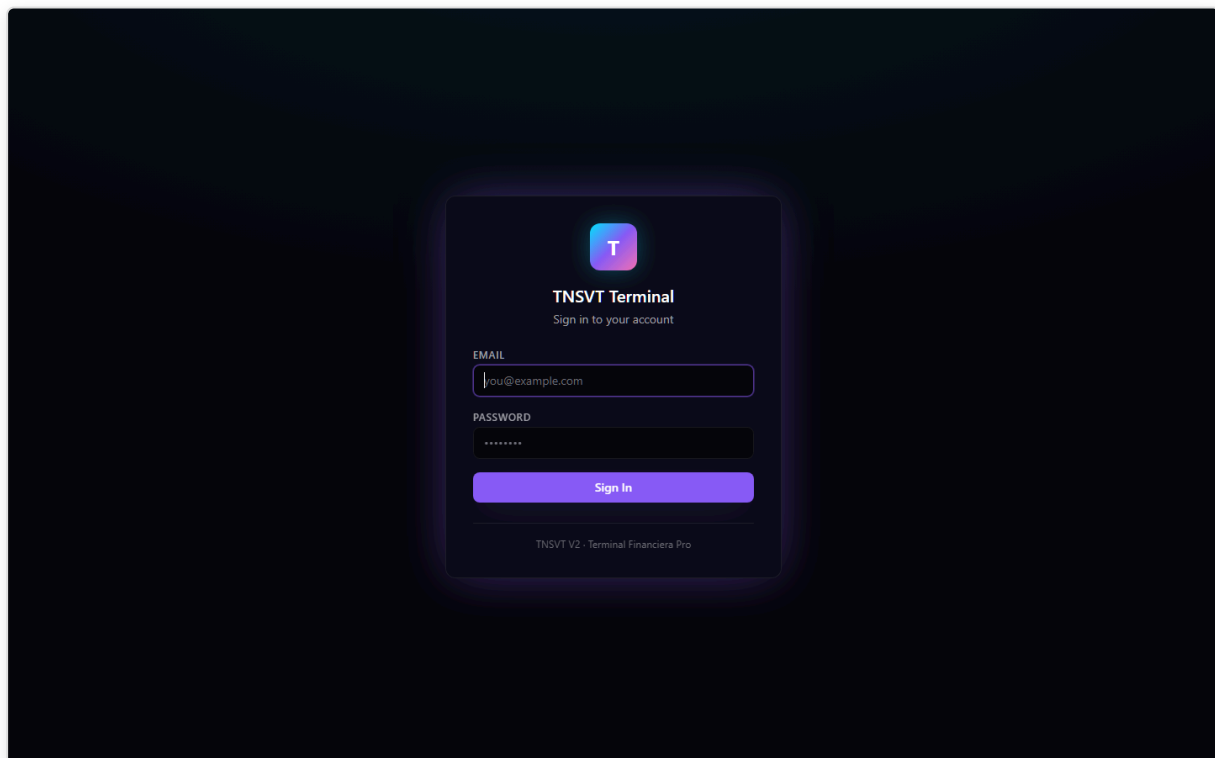
Copy JWT Token

Logout

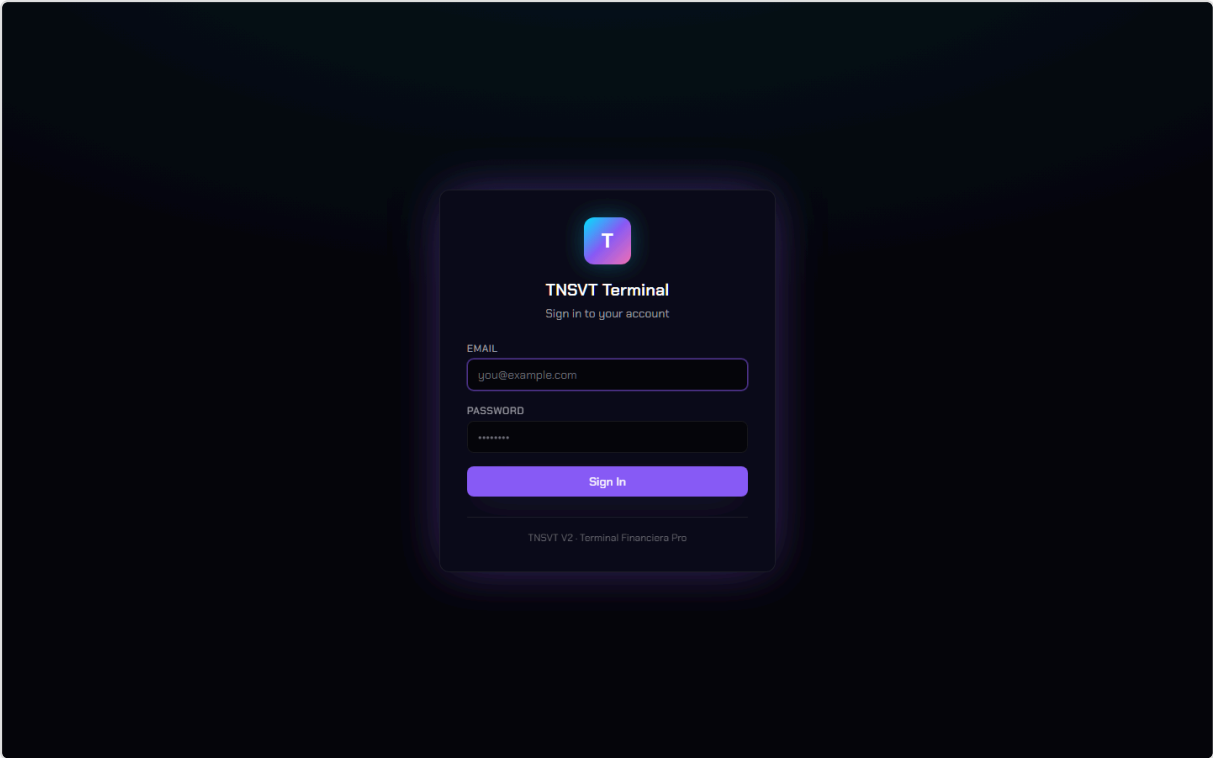
Database Overview (Back office)



Login Screen



Live Dashboard



Live Dashboard (Real-time)

TNSVT
Terminal Financiera

Dashboard

Positions

Signals

Live Ticks

History

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Terminal Financiera - Live

OPEN POSITIONS: 0 UNREALIZED P&L: \$0.00 SUCCESS RATE: +0.0% demo

Dashboard

Total P&L
\$0.00

Win Rate
+0.0%

Open Positions
0

30d Signals
0

Open Positions

No open positions

Recent Signals

No signals yet

Live Prices

SYMBOL	BID	ASK	LAST	SOURCE
BTCUSD	\$1.09	\$1.09	\$1.09	mock
ETHUSD	\$1.09	\$1.09	\$1.09	mock
EURUSD	\$1.09	\$1.09	\$1.09	mock
GBPUSD	\$1.09	\$1.09	\$1.09	mock
USDJPY	\$1.09	\$1.09	\$1.09	mock
XAUUSD	\$1.09	\$1.09	\$1.09	mock

Latest Copy Jobs

No copy jobs yet

Live Page

TNSVT
Terminal Financiera

Dashboard

Positions

Signals

Live Ticks

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Settings

Logout

Terminal Financiera - Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Live TicksERROR6 ticks6 symbols

Latest per symbolUPDATES LIVE

SYMBOL	BID	ASK	MID	LAST	SPREAD	SOURCE	UPDATED
BTCUSD	\$1.09	\$1.09	\$1.09	\$1.09	0.00020	mock	4h ago
ETHUSD	\$1.09	\$1.09	\$1.09	\$1.09	0.00020	mock	4h ago
EURUSD	\$1.09	\$1.09	\$1.09	\$1.09	0.00020	mock	4h ago
GBPUSD	\$1.09	\$1.09	\$1.09	\$1.09	0.00020	mock	4h ago
USDJPY	\$1.09	\$1.09	\$1.09	\$1.09	0.00020	mock	4h ago
XAUUSD	\$1.09	\$1.09	\$1.09	\$1.09	0.00020	mock	4h ago

Raw feed (latest 50)

ETHUSD	bid=1.08500	ask=1.08520	09:00:00
BTCUSD	bid=1.08500	ask=1.08520	09:00:00
XAUUSD	bid=1.08500	ask=1.08520	09:00:00
USDJPY	bid=1.08500	ask=1.08520	09:00:00
GBPUSD	bid=1.08500	ask=1.08520	09:00:00
EURUSD	bid=1.08500	ask=1.08520	09:00:00

E2E Pipeline — Live View

TNSVT
Terminal Financiera

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Terminal Financiera - Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Live TicksOPEN30 ticks5 symbols

Latest per symbolUPDATES LIVE

SYMBOL	BID	ASK	MID	LAST	SPREAD	SOURCE	UPDATED
BTCUSD	\$60,193.52	\$60,199.54	\$60,196.53	\$60,196.53	6.01965	builtin-mock	just now
EURUSD	\$1.09	\$1.09	\$1.09	\$1.09	0.00011	builtin-mock	just now
GBPUSD	\$1.26	\$1.26	\$1.26	\$1.26	0.00013	builtin-mock	just now
USDJPY	\$153.31	\$153.33	\$153.32	\$153.32	0.01533	builtin-mock	just now
XAUUSD	\$2,039.06	\$2,039.26	\$2,039.16	\$2,039.16	0.20392	builtin-mock	just now

Raw feed (latest 50)

BTCUSD	bid=60193.51677	ask=60199.53643	01:35:33
XAUUSD	bid=2039.05553	ask=2039.25944	01:35:33
USDJPY	bid=153.31416	ask=153.32950	01:35:33
GBPUSD	bid=1.26266	ask=1.26279	01:35:33
EURUSD	bid=1.09092	ask=1.09103	01:35:33
BTCUSD	bid=60196.29676	ask=60202.31669	01:35:33
XAUUSD	bid=2038.67567	ask=2038.87955	01:35:33
USDJPY	bid=153.25400	ask=153.26933	01:35:33
GBPUSD	bid=1.26270	ask=1.26282	01:35:33
EURUSD	bid=1.09090	ask=1.09101	01:35:33
BTCUSD	bid=60193.23754	ask=60199.25717	01:35:32
XAUUSD	bid=2039.27653	ask=2039.48047	01:35:32
USDJPY	bid=153.24510	ask=153.26042	01:35:32
GBPUSD	bid=1.26217	ask=1.26230	01:35:32

E2E Pipeline — Final Live

TNSVT
Terminal Financiera

Dashboard

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Terminal Financiera - Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Live TicksOPEN50 ticks5 symbols

Latest per symbol

SYMBOL	BID	ASK	MID	LAST	SPREAD	SOURCE	UPDATED
BTCUSD	\$60,113.11	\$60,119.13	\$60,116.12	\$60,116.12	6.01161	builtin-mock	just now
EURUSD	\$1.10	\$1.10	\$1.10	\$1.10	0.00011	builtin-mock	just now
GBPUSD	\$1.28	\$1.28	\$1.28	\$1.28	0.00013	builtin-mock	just now
USDJPY	\$154.14	\$154.15	\$154.14	\$154.14	0.01541	builtin-mock	just now
XAUUSD	\$2,043.37	\$2,043.58	\$2,043.48	\$2,043.48	0.20435	builtin-mock	just now

Raw feed (latest 50)

BTCUSD	bid=60113.11369	ask=60119.12530	01:42:53
XAUUSD	bid=2043.37348	ask=2043.57783	01:42:53
USDJPY	bid=154.13689	ask=154.15150	01:42:53
GBPUSD	bid=1.27760	ask=1.27773	01:42:53
EURUSD	bid=1.10324	ask=1.10335	01:42:53
BTCUSD	bid=60106.29250	ask=60112.30343	01:42:53
XAUUSD	bid=2043.33842	ask=2043.54277	01:42:53
USDJPY	bid=154.10112	ask=154.11653	01:42:53
GBPUSD	bid=1.27733	ask=1.27745	01:42:53
EURUSD	bid=1.10368	ask=1.10379	01:42:53
BTCUSD	bid=60133.91973	ask=60139.93342	01:42:52
XAUUSD	bid=2043.48798	ask=2043.69234	01:42:52
USDJPY	bid=154.05169	ask=154.06709	01:42:52
GBPUSD	bid=1.27768	ask=1.27781	01:42:52

E2E Final Dashboard

TNSVT
Terminal Financiera

Dashboard

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Terminal Financiera - Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Dashboard

Total P&L\$0.00

Win Rate+0.0%

Open Positions0

30d Signals0

Open Positions

No open positions

Recent Signals

No signals yet

Live Prices

SYMBOL	BID	ASK	LAST	SOURCE
BTCUSD	\$60,102.96	\$60,108.97	\$60,105.97	builtin-mock
EURUSD	\$1.10	\$1.10	\$1.10	builtin-mock
GBPUSD	\$1.28	\$1.28	\$1.28	builtin-mock
USDJPY	\$154.27	\$154.29	\$154.28	builtin-mock
XAUUSD	\$2,042.22	\$2,042.42	\$2,042.32	builtin-mock

Latest Copy Jobs

No copy jobs yet

E2E Dashboard Signals

TNSVT
Terminal Financiera

Dashboard

Positions

Signals

Live Ticks

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Terminal Financiera · Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Dashboard

Total P&L
\$0.00

Win Rate
+0.0%

Open Positions
0

30d Signals
0

Open Positions

No open positions

Recent Signals

No signals yet

Live Prices

STREAMING

SYMBOL	BID	ASK	LAST	SOURCE
BTCUSD	\$60,148.82	\$60,154.84	\$60,151.83	builtin-mock
EURUSD	\$1.09	\$1.09	\$1.09	builtin-mock
GBPUSD	\$1.26	\$1.26	\$1.26	builtin-mock
USDJPY	\$153.27	\$153.28	\$153.28	builtin-mock
XAUUSD	\$2,038.04	\$2,038.24	\$2,038.14	builtin-mock

Latest Copy Jobs

No copy jobs yet

Debug — Home Page

TNSVT
Terminal Financiera

Dashboard

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Terminal Financiera · Live

OPEN POSITIONS0UNREALIZED P&L\$0.00SUCCESS RATE+0.0%demo

Dashboard

Total P&L
\$0.00

Win Rate
+0.0%

Open Positions
0

30d Signals
0

Open Positions

No open positions

Recent Signals

No signals yet

Live Prices

ERROR

Waiting for ticks... (start price-feed)

Latest Copy Jobs

No copy jobs yet